RIO 2017 Logistics Simulation Project

Capacity-Constrained Queue Model Report

Structured project report with JaamSim S0-S2 resource-model results added

Status: R-side data analysis, queue logic, and scenario rationale completed; JaamSim ABC carrier resource model implemented and run for the main capacity scenarios S0-S2. Peak-day policy scenarios S3-S4 remain R-side operating-policy extensions until weekday-specific JaamSim capacity logic is added.

Executive Summary

This report documents the current stage of the RIO 2017 logistics simulation project. The original prepared dataset contained 2,548 orders. After removing one missing interarrival record and two negative carrier-dispatch records, 2,545 valid orders were used for R-side analysis. Input distributions were fitted for interarrival time, order approval time, carrier handoff/dispatch time, and delivery transit time. Because the dataset does not include observed carrier capacities, the carrier-capacity queue is introduced as a scenario-based operational assumption rather than a measured real-world capacity system. The base capacity scenario uses daily carrier capacities of 3, 4, and 5 orders for Carrier_A, Carrier_B, and Carrier_C, respectively. Under the R-side base queue scenario, 693 orders experience queue delay, corresponding to a delay rate of 27.23%. The main capacity scenarios were then implemented in JaamSim using an ABC carrier resource structure with queues, seize/release logic, and carrier resources. For 30-replication JaamSim runs, the mean total system time was 15.25 days for S0, 15.10 days for S1, and 15.06 days for S2. These results are moderately higher than the R-side benchmark means, but they confirm the expected operational direction: increasing carrier capacity reduces the mean total system time.

Methodological status note

This report includes the completed R-side data analysis, queue-rule design, scenario rationale, and the implemented JaamSim ABC carrier resource model for S0, S1, and S2. The JaamSim model uses explicit queues, seize/release logic, and carrier resources for Carrier_A, Carrier_B, and Carrier_C. S3 and S4 are retained as R-side operating-policy extensions because their peak-day overtime logic requires weekday-specific capacity changes that were not implemented in the current JaamSim model. Therefore, final conclusions distinguish between JaamSim-implemented capacity scenarios (S0-S2) and R-side policy scenarios (S3-S4).

1. Problem Formulation

1.1 Orientation

The system studied in this project is an e-commerce logistics process in which customer orders are received, approved, handed over to a carrier, and finally delivered to the customer. Each order is treated as an entity moving through a sequence of time-consuming process stages. The original data provide timestamps and calculated durations for the order life cycle, but the data do not directly include carrier capacity or staffing information. Therefore, the capacity-constrained queue mechanism is introduced to represent a plausible operational bottleneck at the carrier handoff stage.

The physical interpretation of the model is that orders arrive over time, wait for approval, wait to be handed over to a carrier, and then enter the delivery transit stage. On busy carrier handoff days, the number of orders assigned to a carrier profile can exceed its assumed daily capacity. In that case, excess orders experience queue delay before entering the delivery transit stage.

1.2 Problem Statement

The main problem is to evaluate how a capacity-constrained carrier handoff process affects total delivery performance. The dataset contains the main order-processing and delivery durations, but it does not provide actual carrier capacity values. For this reason, the project does not claim to estimate real carrier capacities. Instead, it uses scenario-based capacity values to study queue formation and alternative capacity policies.

1.3 Objectives

- Clean and prepare the RIO 2017 logistics data for simulation modeling.
- Identify and fit input distributions for the main process-time variables.
- Develop a capacity-constrained queue mechanism for the carrier handoff stage.
- Compare alternative carrier-capacity and peak-day overtime scenarios.
- Implement the main S0-S2 carrier-capacity scenarios in JaamSim using explicit queue, seize/release, and resource logic.
- Generate recommendations based on queue delay, total system time, and scenario performance.

2. System Definition

2.1 Identify System Components

Table 1. Main system components.

Component	Model role
Order entity	A customer order moving through the system.
Entity Generator	Creates order arrivals using fitted interarrival times.
Order Approval Delay	Represents the time between purchase and approval.
Carrier Profile Assignment	Assigns each order to Carrier A, Carrier B, or Carrier C.
Approval-to-Carrier Handoff Delay	Represents the time from approval to carrier handoff.
Carrier Capacity / Queue	Checks daily capacity and adds queue delay to excess orders.
Delivery Transit Delay	Represents delivery transit after carrier handoff.
Customer Delivery / System Exit	Records completed orders and total system time.

2.2 Identify Input and Output Variables

Table 2. Main input, scenario, and output variables.

Variable	Type	Unit
order purchase interval / arrival min	Input	Minutes
order approval time / approval min	Input	Minutes
carrier dispatch time / dispatch days	Input	Days
delivery transit time / transit days	Input	Days
carrier id	Scenario parameter	Profile label
carrier capacity	Scenario parameter	Orders/day
queue delay days	Output	Days
total system time with queue days	Output	Days

Note: Carrier_A, Carrier_B, and Carrier_C are scenario carrier profiles, not real company names.

2.3 Key Modeling Assumptions

Table 2A. Key modeling assumptions used in the simulation model.

Assumption	Explanation
Single entity type	All orders are modeled as one order type. The model does not distinguish different customer or product classes.
Scenario carrier profiles	Carrier_A, Carrier_B, and Carrier_C are scenario carrier profiles, not observed real company names.
Carrier routing	Orders are routed to Carrier_A, Carrier_B, or Carrier_C using observed carrier-profile proportions from the cleaned dataset.
FIFO carrier queues	Orders waiting for carrier service follow first-in-first-out queue discipline.
Unlimited waiting space	Carrier queues do not reject orders; orders wait until carrier resource capacity is available.
Carrier capacity scenarios	S0 uses A=3, B=4, C=5; S1 uses A=4, B=5, C=6; S2 uses A=5, B=6, C=7 orders/day.
Terminating simulation	The JaamSim model represents a fixed yearly operating horizon rather than an infinite steady-state system.
No warm-up period	Each replication starts empty, so no warm-up period is used.
Carrier service stage	The carrier service stage represents one-day handoff capacity processing before delivery transit begins.

3. Input Data Collection

3.1 Data Collection Process

The dataset consists of order-level logistics records from the RIO 2017 data file. The relevant fields include order purchase timestamps, order approval timestamps, carrier handoff timestamps, customer delivery timestamps, and derived process durations. These fields were used to compute interarrival time, approval time, carrier handoff/dispatch time, and delivery transit time.

The raw prepared file contained 2,548 rows. After cleaning, 2,545 observations were retained. Three rows were removed: one due to a missing interarrival value and two due to negative carrier dispatch times. Negative process durations are physically impossible and were therefore excluded before total-system-time and scenario calculations.

Table 3. Data quality summary.

Variable	N total	Missing	Zero	Negative	Positive
Interarrival Time - Minutes	2548	1	12	0	2535
Order Approval Time - Minutes	2548	0	82	0	2466
Order Approval Time - Days	2548	0	82	0	2466
Carrier Dispatch Time - Days	2548	0	0	2	2546
Delivery Transit Time - Days	2548	0	2	0	2546

Note: Queue-delay, scenario, and statistical results are based on the cleaned dataset with n = 2,545 valid orders unless otherwise stated. The weekday operation-pattern table is reported from the prepared Excel evidence sheet with n = 2,548 to keep the handoff counts consistent with the source workbook.

3.2 Data Analysis Process

The dataset uses mixed time units. Interarrival time and order approval time are measured in minutes, whereas carrier dispatch time and delivery transit time are measured in days. Therefore, order approval time was divided by 1440 before computing total system time. The historical no-queue total system time was calculated as approval days + dispatch days + transit days.

Table 4. Descriptive statistics for main input variables.

Variable	Unit	N	Mean	Median	SD	Minimum	Maximum	Outliers
Interarrival Time	Minutes	2545	201.58	86.75	331.05	0	5799.68	324
Order Approval Time	Minutes	2545	218.13	14.18	673.94	0	7771.75	528
Carrier Dispatch Time	Days	2545	3.09	2.11	3.47	0.01	40.34	136
Delivery Transit Time	Days	2545	10.58	8.15	9.15	0	162.29	163

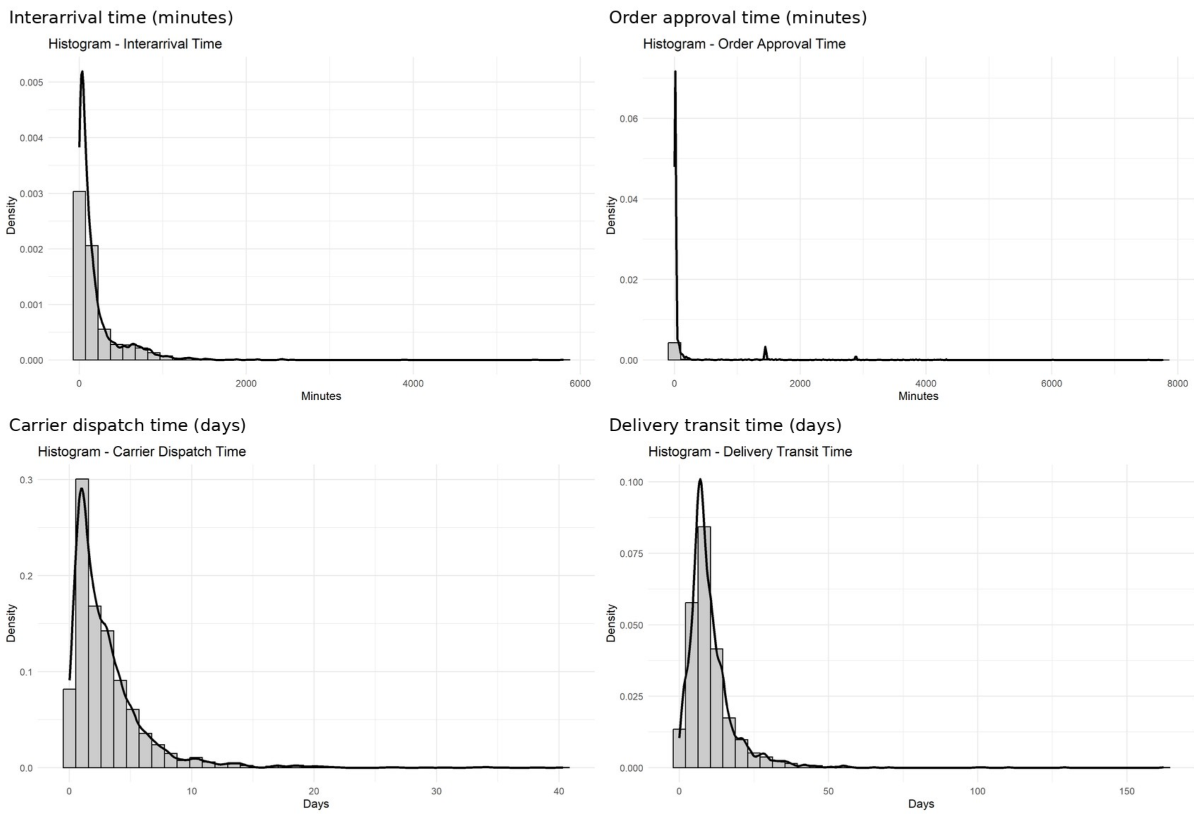

Figure 1. Histograms and density curves for the main input variables.

Candidate distributions were fitted by maximum likelihood estimation. The tested candidates were Normal, Exponential, Gamma, Lognormal, and Weibull. Distribution selection was based mainly on AIC, with BIC, KS statistic, and graphical diagnostics used as supporting evidence. P-values were not used as the only decision criterion because large samples can reject even small practical deviations from theoretical distributions.

Table 5. AIC comparison for input distribution fitting.

Variable	Normal	Exponential	Gamma	Lognormal	Weibull	Selected
Interarrival Time	36593.21	31973.02	31433.31	31769.03	31348.60	Weibull
Order Approval Time	39148.50	31616.19	Fit failed	25634.81	27069.49	Lognormal
Carrier Dispatch Time	13560.54	10827.89	10759.64	10709.08	10807.06	Lognormal
Delivery Transit Time	18475.45	17090.31	16409.45	16393.46	16633.91	Lognormal

Table 6. Selected input distributions and estimated parameters.

Variable	Distribution	Parameters	Unit
Interarrival Time	Weibull	shape = 0.709632; scale = 159.926387	Minutes
Order Approval Time	Lognormal	meanlog = 3.326189; sdlog = 1.580974	Minutes
Carrier Dispatch Time	Lognormal	meanlog = 0.673256; sdlog = 1.011025	Days
Delivery Transit Time	Lognormal	meanlog = 2.106766; sdlog = 0.738423	Days

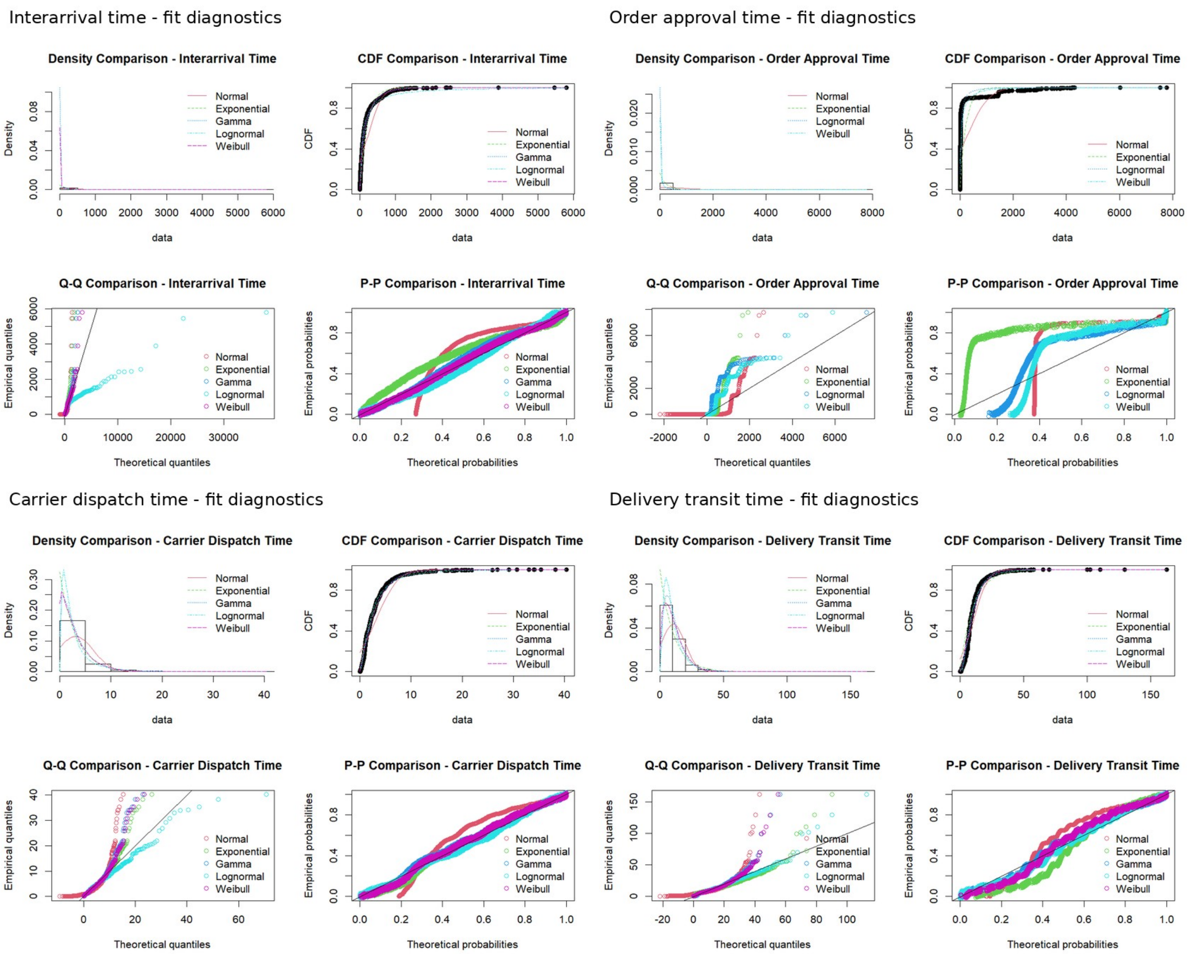

Figure 2. Distribution fit diagnostic plots for the main input variables.

4. Model Translation

4.1 Flow Chart of Model

Figure 3 shows the conceptual logic of the simulation model. It is not a JaamSim screenshot; it is the model-flow chart required before translating the logic into JaamSim objects.

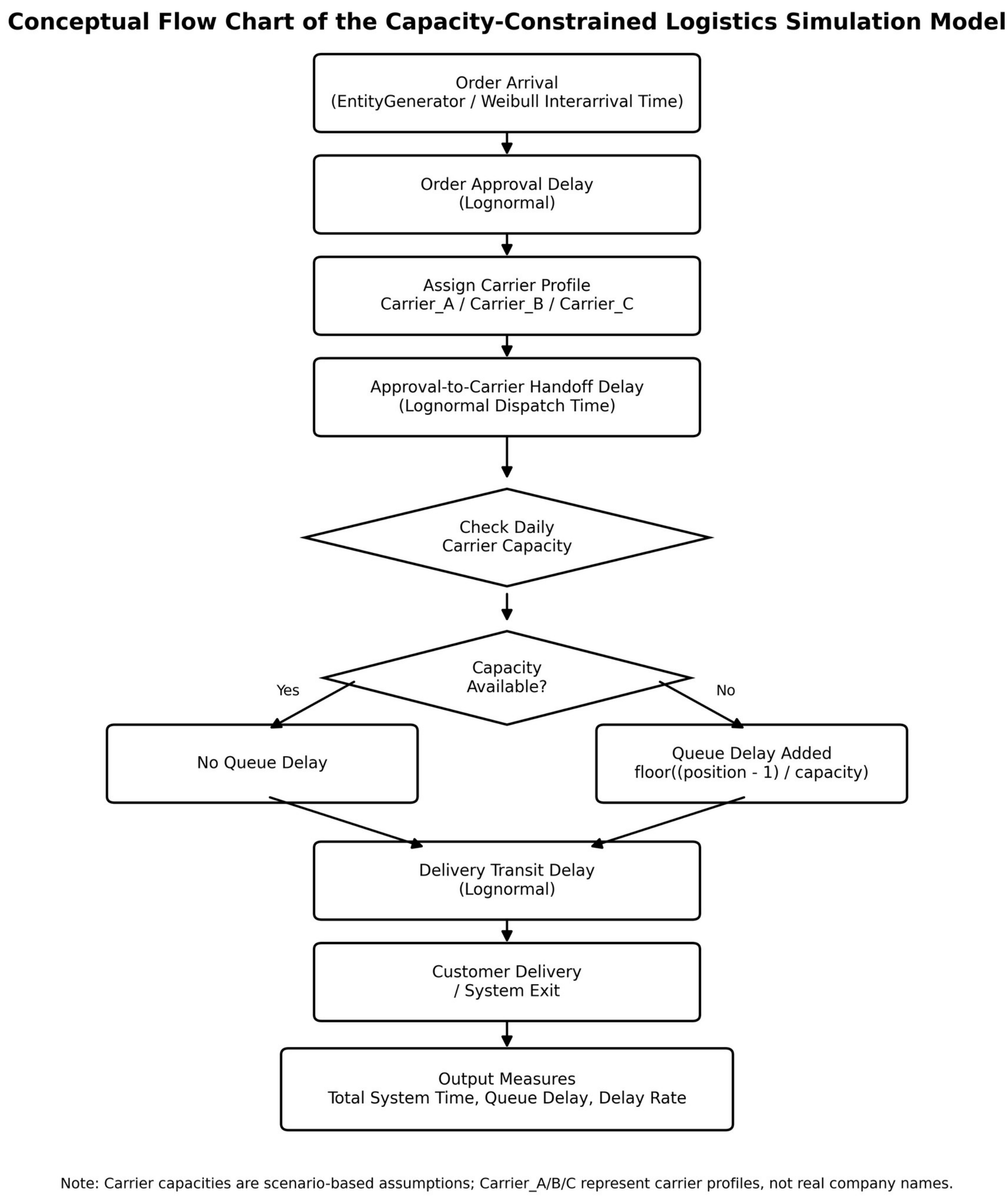

Figure 3. Conceptual flow chart of the capacity-constrained logistics simulation model.

4.2 Developing Model in JAAMSIM

The main JaamSim implementation has been completed for the capacity scenarios S0, S1, and S2. The model translates the conceptual flow into an ABC carrier-resource structure: orders are generated, pass through approval and dispatch delays, are routed to Carrier_A, Carrier_B, or Carrier_C, wait in the corresponding carrier queue if the carrier resource is unavailable, receive carrier service through seize/release logic, pass through delivery transit delay, and exit the system. The same fitted arrival, approval, dispatch, and transit distributions were used. The carrier capacities were set to A=3, B=4, C=5 for S0; A=4, B=5, C=6 for S1; and A=5, B=6, C=7 for S2. The carrier branch uses routing probabilities based on observed carrier-profile proportions in the cleaned dataset: Carrier_A = 0.2908, Carrier_B = 0.4373, and Carrier_C = 0.2719. A screenshot of the implemented JaamSim model layout is provided below.

JaamSim component	Project meaning
EntityGenerator	Generates customer orders using the fitted Weibull interarrival-time distribution.
Approval Delay	Represents order approval time using the fitted Lognormal distribution.
Dispatch/Handoff Delay	Represents approval-to-carrier handoff time using the fitted Lognormal distribution.
Carrier Branch	Routes orders to Carrier_A, Carrier_B, or Carrier_C using observed carrier-profile proportions: Carrier_A = 0.2908, Carrier_B = 0.4373, Carrier_C = 0.2719.
Carrier Queues	Represent waiting areas before each carrier resource. Waiting space is treated as unlimited.
Carrier Resources	Represent daily carrier handoff capacity. S0 uses A=3, B=4, C=5; S1 uses A=4, B=5, C=6; S2 uses A=5, B=6, C=7.
Seize / Service / Release logic	Seizes the assigned carrier resource, processes the handoff service, and releases the resource after completion.
Transit Delay	Represents delivery transit time using the fitted Lognormal distribution.
EntitySink / Statistics	Records completed orders and total system time.

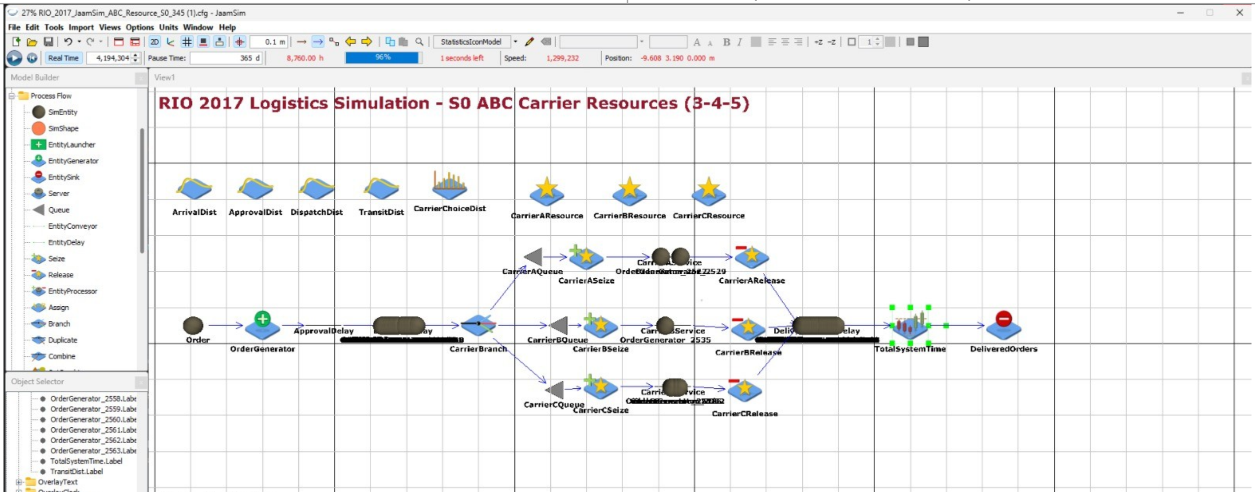

Figure 3A. Implemented JaamSim ABC carrier-resource model layout for the main capacity scenarios.

5. Verification

5.1 Debug

R-side verification was performed for the data preparation and queue-rule logic, and JaamSim debugging was completed for the main S0-S2 carrier resource model. The queue-delay formula was verified in the R/Excel evidence file using floor((queue_position - 1) / carrier_capacity). In JaamSim, the implemented model was checked to ensure that orders moved through arrival, approval delay, dispatch delay, carrier branch, carrier queue, carrier resource service, transit delay, statistics, and system exit without negative processing times or blocked entities.

- Check that orders move through the sequence: arrival -> approval -> handoff -> queue/capacity check -> transit -> system exit.
- Check that no order receives negative process time.
- Check that carrier resource capacities change correctly across S0, S1, and S2.
- Check that orders wait in the correct Carrier_A, Carrier_B, or Carrier_C queue when the corresponding carrier resource is busy.

- Check that total system time equals approval days + dispatch days + queue delay days + transit days.

5.2 Animate

Animation verification was performed after the JaamSim model was built. During the animation, orders were observed entering from the generator, passing through approval and dispatch delays, being routed to the A/B/C carrier queues, seizing and releasing the corresponding carrier resources, moving through the delivery transit delay, and exiting through the sink. The screenshot below shows the implemented resource-based carrier handoff structure during model execution.

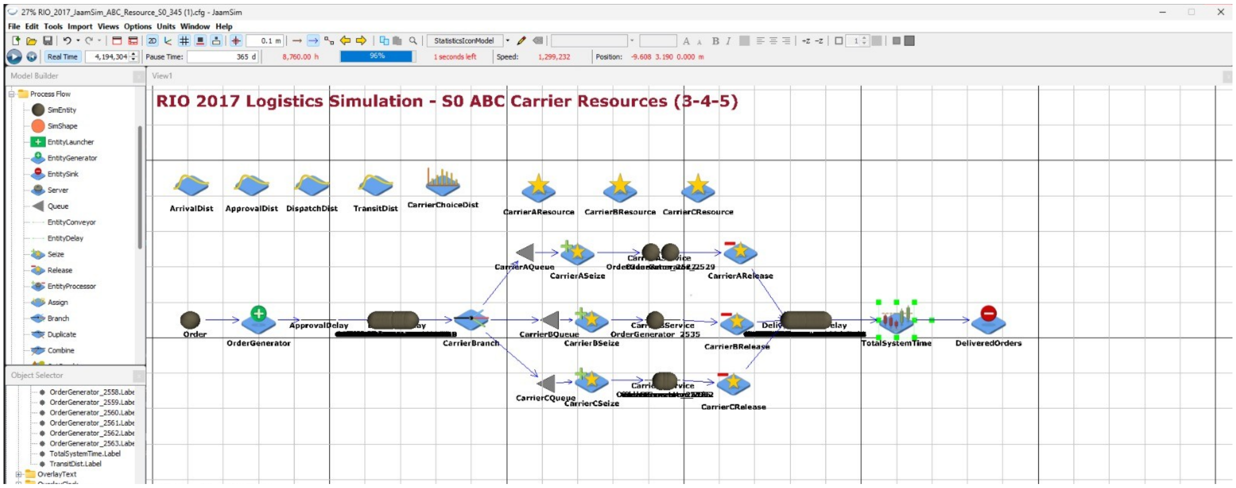

Figure 3B. Animation verification screenshot showing A/B/C carrier queues, resources, and entity movement.

6. Validation

6.1 Face Validity

Face validity is partially supported by the structure of the model. Orders follow the real logistical sequence of purchase, approval, carrier handoff, and delivery. The capacity mechanism is also operationally plausible because carrier handoff operations can be constrained on busy days. However, the carrier capacity values are scenario assumptions and should not be interpreted as observed real capacities.

The base 3-4-5 scenario creates a visible but not catastrophic queue: 693 of 2,545 cleaned orders experience delay, and most delays are one day. This supports the face validity of the scenario as a constrained-capacity what-if model rather than an extreme system-collapse case.

6.2 Statistical Validity and Benchmark Comparison

For the main JaamSim capacity scenarios S0, S1, and S2, validation was performed as a benchmark comparison rather than as a full hypothesis-test-based validation. Each scenario was run with 30 replications and a 365-day run duration with no warm-up period. The JaamSim mean total system time was compared with the corresponding R-side benchmark mean, and the validation error was calculated as abs(JaamSim mean - R mean) / R mean x 100. Because replication-level JaamSim outputs were not exported separately, a formal test such as a two-sample t-test or nonparametric rank-sum test between R-side and JaamSim replication outputs was not performed. Therefore, the validation evidence focuses on relative error and on whether the JaamSim implementation preserves the expected scenario direction. S3 and S4 were not implemented in JaamSim at this stage because their peak-day overtime logic requires weekday-specific capacity changes; they remain R-side policy scenarios.

Table 8. JaamSim benchmark validation results for main capacity scenarios.

Note: S0-S2 were implemented in JaamSim with explicit A/B/C queue-resource logic. The table is interpreted as benchmark validation rather than a formal hypothesis test because replication-level JaamSim outputs were not exported separately. S3-S4 remain R-side operating-policy scenarios until weekday-specific JaamSim overtime logic is added.

Scenario	R mean days	R SD days	R mean hours	JaamSim mean days	Error (%)	Decision
S0 Base Capacity 345	14.18	10.05	340.29	15.25	7.59	Validated direction; moderate deviation
S1 Mild Capacity 456	14.05	10.06	337.15	15.10	7.48	Validated direction; moderate deviation
S2 Stronger Capacity 567	13.97	10.06	335.32	15.06	7.77	Validated direction; moderate deviation

Output	Value
SoftwareName	"JaamSim"
SoftwareVersion	"2026-03"
ConfigurationFile	"C:\Users\msi-nb\Downloads\RIO_2017_JaamSim_ABC_Resource_S1_456 (1).cfg"
ScenarioNumber	1
ScenarioIndex	{1}
ReplicationNumber	30
RunNumber	30
RunIndex	{1}
PresentTimeAndDate	"Jun 04, 2026 20:24"
PresentSimulationTime	8760.00[h]
SimDate	{1971, 1, 1, 0, 0, 0, 0}
SimDayOfWeek	6

Figure 3C. JaamSim output screen confirming 30 replications for the S1 scenario.

7. Experimental Design

7.1 Alternative Configurations

A formal factorial experimental design was not applied at this stage. Instead, scenario-based what-if analysis was used because the carrier capacities are not observed historical company capacities; they are operational policy assumptions added to the RIO dataset to study queue behavior. The base scenario, S0_Base_Capacity_345, represents the constrained carrier-handoff system. The other scenarios are not intended to prove that higher capacity is mechanically better. Higher capacity is expected to reduce queueing. The purpose is to quantify how much improvement is obtained from each capacity policy and to compare permanent capacity expansion with peak-day overtime alternatives.

Therefore, the experimental question is: how much total-system-time and queue-delay reduction is gained by moving from 3-4-5 capacity to 4-5-6 or 5-6-7, and can a more realistic peak-day support policy provide similar benefits without permanently increasing capacity every day? This framing makes the scenario analysis useful for decision making rather than simply ranking the largest capacity as the best option.

Design-control note for S4: S4_Balanced_456_Peak_Overtime intentionally combines two policy changes: a moderate permanent capacity increase from 3-4-5 to 4-5-6 and additional peak-day overtime. This mixed policy is included as a realistic managerial alternative, not as a pure one-factor level. To avoid confusing the source of the improvement, S4 is interpreted together with S1 and S3: S1 isolates the effect of the 4-5-6 capacity increase, S3 isolates the effect of peak-day overtime under the original 3-4-5 capacity, and S4 shows the combined policy effect. Therefore, S4 should not be used alone to claim which single factor caused the improvement; it is used to evaluate whether the combined policy is operationally attractive.

Table 9. Scenario definitions.

Scenario	Definition
Benchmark Historical No Queue	Reference only; original total system time without queue.
S0 Base Capacity 3-4-5	Base queue model with capacities A=3, B=4, C=5.
S1 Mild Capacity 4-5-6	Mild capacity increase to A=4, B=5, C=6.
S2 Stronger Capacity 5-6-7	Stronger capacity increase to A=5, B=6, C=7.
S3 Peak-Day Overtime 3-4-5	Base capacities plus extra Monday/Tuesday capacity.
S4 Balanced 4-5-6 + Peak Overtime	Mild capacity plus limited Monday/Tuesday overtime capacity.

7.2 Conduct Production Simulation Runs

JaamSim production simulation runs were completed for the main capacity scenarios S0, S1, and S2. Each scenario was run for 30 independent replications with a run duration of 365 days and no warm-up period. The system was treated as a terminating simulation: each replication started with an empty system. The output collected from each scenario was the mean total system time in hours, converted to days for comparison with the R-side benchmark. S3 and S4 were not run in JaamSim because their peak-day overtime logic requires weekday-specific resource-capacity schedules; these scenarios remain R-side operating-policy extensions. Thirty replications were used as a practical replication count to reduce stochastic variation; a formal sequential stopping procedure was not applied.

Note: Sections 7 and 8 now combine R-side scenario analysis with JaamSim production-run results for S0-S2. S3 and S4 should still be interpreted as R-side policy scenarios until peak-day overtime is implemented in JaamSim.

8. Analysis

Important interpretation note: The numerical comparisons in this section combine R-generated scenario outputs with JaamSim production results for S0-S2. The S0-S2 JaamSim model uses explicit A/B/C carrier queues, seize/release logic, and carrier resources. Therefore, JaamSim means are expected to differ moderately from R-side deterministic benchmark means, but the key validation criterion is whether the implemented model preserves the scenario direction and operational interpretation.

8.1 Queue and Congestion Analysis

For the base 3-4-5 capacity scenario, queue delay was analyzed as an empirical discrete distribution. This is more appropriate than forcing a continuous distribution because queue_delay_days is generated by a deterministic capacity rule and takes integer values such as 0, 1, 2, 3, 4, and 5 days.

Table 10. Empirical distribution of queue delay under S0 base capacity 3-4-5.

Queue delay (days)	Orders	Percent (%)
0	1852	72.77
1	527	20.71
2	126	4.95
3	27	1.06
4	9	0.35
5	4	0.16

Note: The values in Table 10 are based on the cleaned dataset with n = 2,545. The prepared Excel file contains 2,548 rows before cleaning.

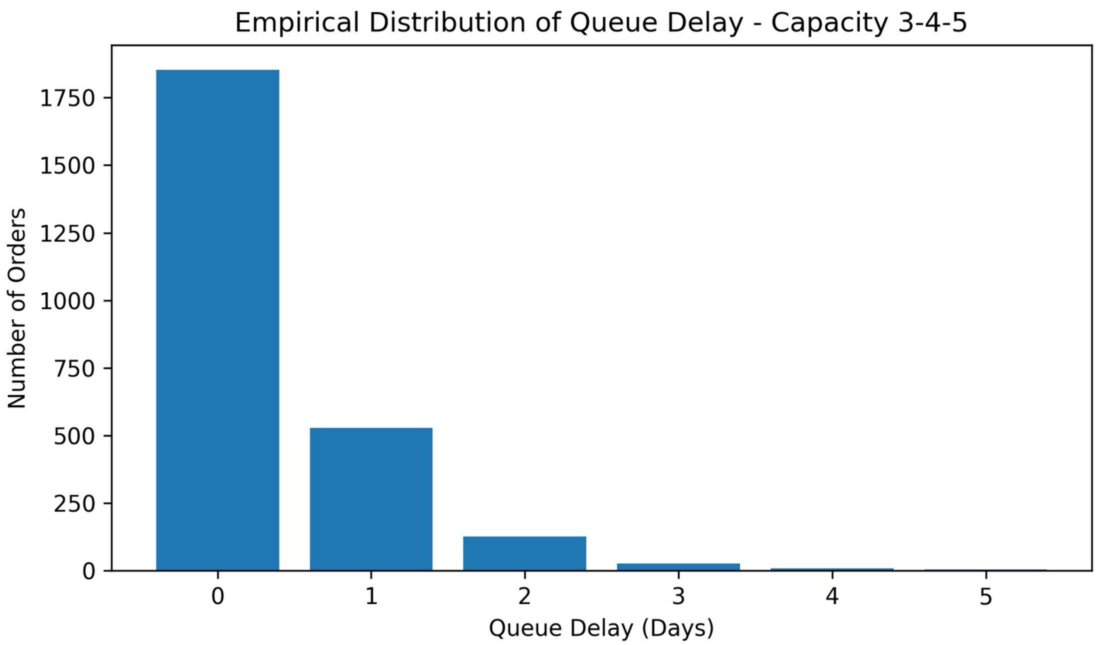

Figure 4. Empirical queue-delay distribution under S0 base capacity 3-4-5.

Table 11. Weekday-based handoff and queue-delay summary.

Weekday	Handoffs	Delayed orders	Delay rate (%)	Mean queue delay	Max delay
Monday	576	188	32.64	0.46	4
Tuesday	540	157	29.07	0.43	5
Wednesday	492	143	29.07	0.36	3
Thursday	454	110	24.23	0.29	3
Friday	453	95	20.97	0.25	3
Saturday	33	0	0.00	0.00	0
Sunday	0	0	0.00	0.00	0

Note: Table 11 uses the prepared Excel Weekday Analysis sheet with n = 2,548 rows to show the full weekday handoff pattern. Core scenario statistics, distribution fitting, and hypothesis tests remain based on the cleaned dataset with n = 2,545 valid orders. Saturday handoffs are limited and no Sunday handoff appears in the prepared data.

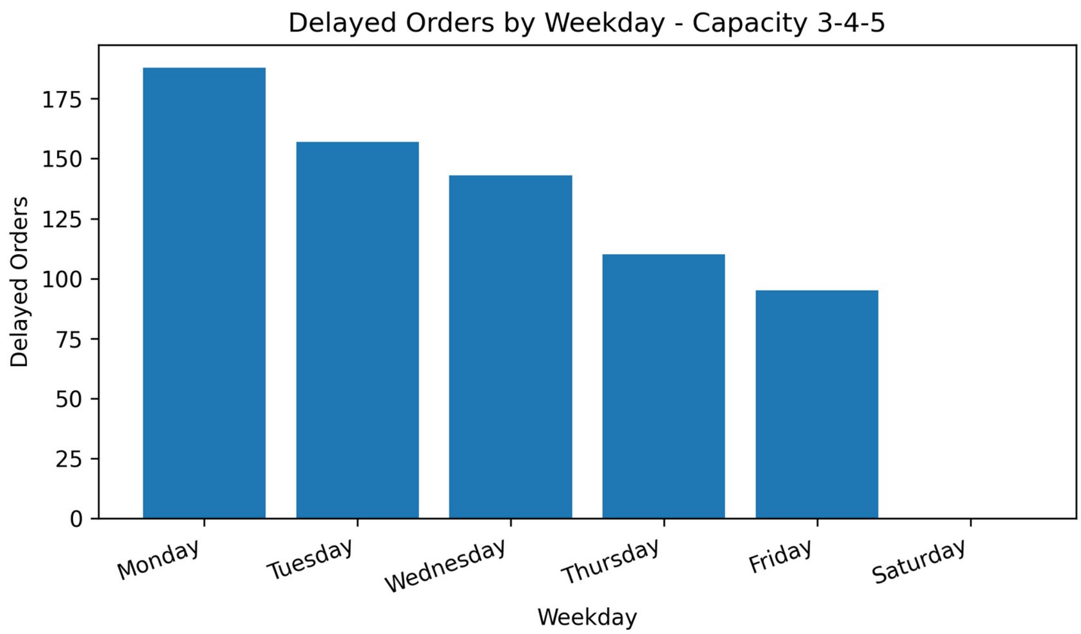

Figure 5. Delayed orders by weekday under the base 3-4-5 capacity scenario.

Table 12. Top congested real carrier handoff dates under S0.

Date	Weekday	Total load	Overloaded carriers	Overloaded load	Overloaded capacity	Delayed orders	Max delay
2017-11-27	Monday	50	Carrier A, Carrier B, Carrier C	50	12	38	4
2017-11-28	Tuesday	45	Carrier B, Carrier C	42	9	33	5
2017-11-29	Wednesday	35	Carrier A, Carrier B, Carrier C	35	12	23	3
2017-11-30	Thursday	35	Carrier A, Carrier B, Carrier C	35	12	23	3
2017-12-05	Tuesday	29	Carrier B, Carrier C	26	9	17	3
2017-12-08	Friday	26	Carrier B, Carrier C	23	9	14	3
2017-12-04	Monday	25	Carrier A, Carrier B, Carrier C	25	12	13	2
2017-12-06	Wednesday	25	Carrier A, Carrier B, Carrier C	25	12	13	3

Note: For 2017-11-28, total daily load is 45. The overloaded-carrier load is 42 because Carrier_A had load 3 and did not exceed its capacity; only Carrier_B and Carrier_C were overloaded.

8.2 Scenario Comparison and Statistical Tests

The scenario comparison should be interpreted as a marginal-benefit analysis, not as a claim that the highest capacity is unexpectedly better. Since S1 and S2 increase the daily capacities, their queue delays and mean total system times are expected to improve relative to S0. The important result is the size of the improvement and whether a targeted peak-day support policy can provide a practically meaningful alternative to permanent capacity expansion. In this project, capacity expansion reduces delay, but the improvement is moderate because delivery transit time remains the dominant component of total system time.

Table 13. Scenario summary for total system time.

Scenario	Role	Mean days	Median days	SD days	Mean hours	Improvement vs S0 (days)	Improvement vs S0 (%)
Benchmark Historical No Queue	Reference benchmark	13.82	11.57	10.03	331.65	0.36	2.54
S0 Base Capacity 345	Experimental scenario	14.18	11.99	10.05	340.29	0	0
S1 Mild Capacity 456	Experimental scenario	14.05	11.91	10.06	337.15	0.13	0.92
S2 Stronger Capacity 567	Experimental scenario	13.97	11.83	10.06	335.32	0.21	1.46
S3 PeakDay Overtime 345	Experimental scenario	14.07	11.90	10.05	337.75	0.11	0.75
S4 Balanced 456 Peak Overtime	Experimental scenario	14.01	11.84	10.06	336.22	0.17	1.19

Table 14. Queue KPI summary by experimental scenario.

Scenario	Delayed orders	Delay rate (%)	Mean queue delay	Max delay
S0 Base Capacity 345	693	27.23	0.36	5
S1 Mild Capacity 456	483	18.98	0.23	4
S2 Stronger Capacity 567	335	13.16	0.15	3
S3 PeakDay Overtime 345	531	20.86	0.25	3

S4 Balanced 456 Peak Overtime	415	16.31	0.19	3

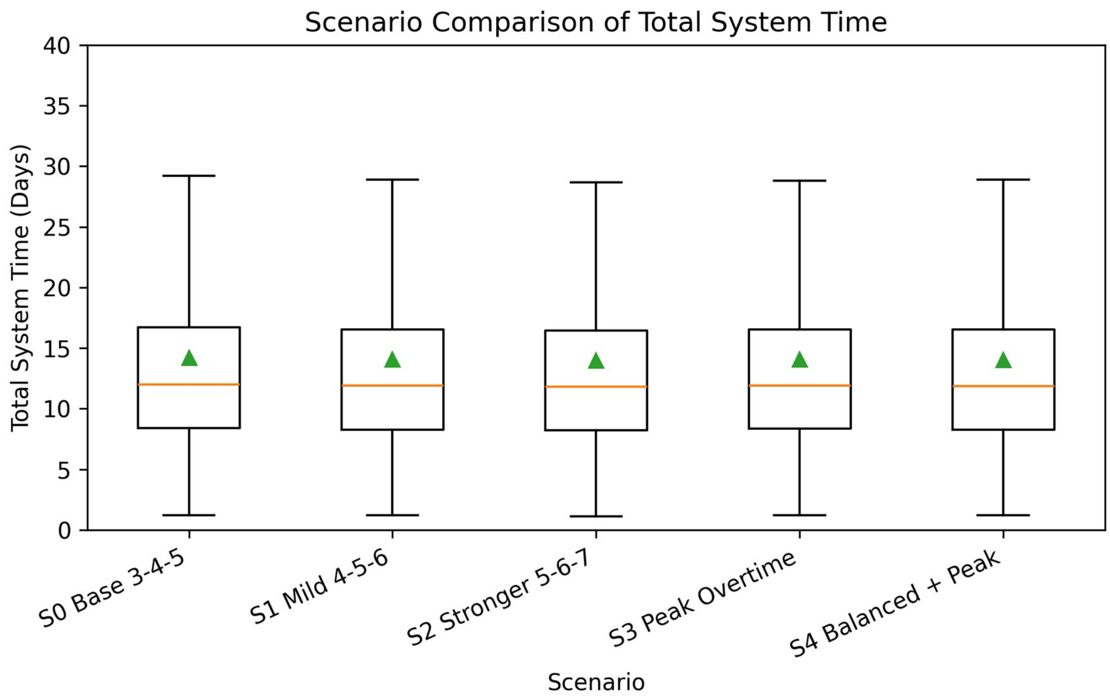

Figure 6. Scenario comparison boxplot for total system time.

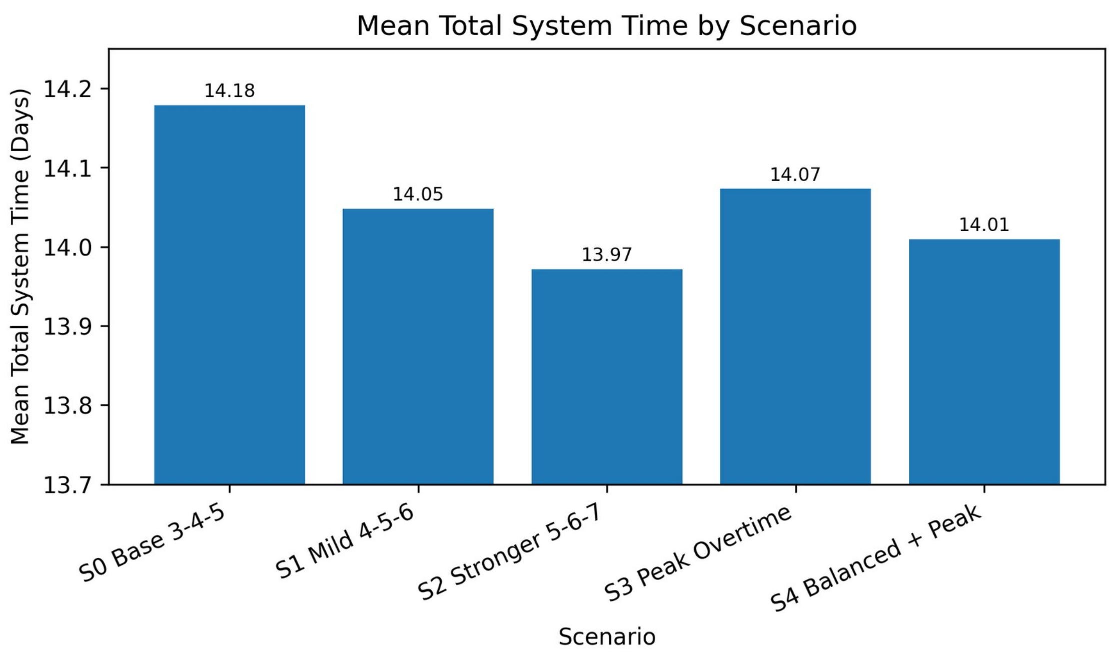

Figure 7. Mean total system time by scenario.

S2_Stronger_Capacity_567 gives the lowest mean total system time among the experimental scenarios, so it is the best performance-oriented alternative. However, this result should not be interpreted as a surprising finding that larger capacity is better; it is the expected direction. The meaningful interpretation is that moving from S0 3-4-5 to S2 5-6-7 improves the mean total system time by about 0.21 days, or approximately 5.0 hours per order. S1 4-5-6 improves the mean by about 0.13 days, while S4 4-5-6 plus peak-day overtime improves it by about 0.17 days. Therefore, S4 may be more operationally realistic if permanent 5-6-7 capacity is costly, because it captures much of the benefit of capacity expansion while focusing extra support on high-load days.

Separation of S4 effects: The improvement of S4 should be read relative to the single-policy scenarios. S1, which changes only capacity from 3-4-5 to 4-5-6, improves the mean total system time by about 0.13 days. S3, which keeps the base 3-4-5 capacity but adds peak-day overtime, improves the mean by about 0.11 days. S4 combines these two ideas and improves the mean by about 0.17 days. This combined result is therefore expected, but it is still useful because it shows that the two policies together provide a larger improvement than either mild capacity expansion or peak-day overtime alone, while remaining less aggressive than permanent 5-6-7 capacity.

The practical conclusion is that capacity management matters, but it is not the only bottleneck in the logistics process. Queue delay is reduced by capacity policies, whereas total delivery performance is still strongly affected by the delivery transit-time distribution. For this reason, the recommendation should balance performance gain, operational feasibility, and potential cost rather than selecting the maximum-capacity scenario automatically.

JaamSim production-run interpretation: The 30-replication JaamSim results for S0-S2 show the same scenario direction as the R-side benchmark. Mean total system time decreased from 15.25 days in S0 to 15.10 days in S1 and 15.06 days in S2. The validation errors relative to the R-side benchmark means are approximately 7.59%, 7.48%, and 7.77%, respectively. These errors are moderate but explainable because the JaamSim model implements the carrier handoff stage with explicit queue, resource, seize, service, and release logic, while the R-side model uses a deterministic daily capacity-rule calculation. The important validation result is that both approaches agree on the main operational direction: increasing carrier capacity reduces mean total system time.

Table 15. One-way ANOVA for experimental capacity scenarios.

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Scenario	4	62.88	15.72	0.16	0.96
Residuals	12720	1286376.22	101.13		

The ordinary one-way ANOVA gives F = 0.155 and p = 0.961, so scenario means are not strongly separated under an independent-sample framing. This result should be interpreted cautiously for two reasons: first, all scenario outputs are generated from the same order records; second, these are R-side scenario outputs rather than final JaamSim replication results. The ANOVA table is therefore used as an exploratory comparison, while paired comparisons and practical mean differences are more informative for this stage.

Table 16. Paired t-tests against the base S0 queue scenario.

Comparison	t statistic	DF	p-value	Mean diff. days	CI lower	CI upper
S0 vs S1 Mild Capacity 4-5-6	19.57	2544	1.48e-79	0.13	0.12	0.14
S0 vs S2 Stronger Capacity 5-6-7	25.24	2544	1.31e-125	0.21	0.19	0.22
S0 vs S3 Peak-Day Overtime 3-4-5	16.73	2544	1.17e-59	0.11	0.09	0.12
S0 vs S4 Balanced 4-5-6 + Peak Overtime	22.22	2544	3.87e-100	0.17	0.15	0.18

Paired comparisons are more informative because each scenario is generated from the same orders. These tests show statistically significant mean reductions compared with S0, but the practical improvements remain moderate.

9. Recommendations and Conclusions

9.1 Recommendations

- Use S0_Base_Capacity_345 as the base capacity-constrained queue scenario for JaamSim implementation and validation. This is the main reference point for measuring how much the queue is reduced by alternative capacity policies.
- Use S2_Stronger_Capacity_567 as the best performance-oriented scenario because it gives the lowest mean total system time and the lowest queue impact among the tested alternatives.
- Consider S4_Balanced_456_Peak_Overtime as the more operationally realistic recommendation if permanent 5-6-7 capacity is costly. It provides a stronger improvement than simple 4-5-6 capacity while concentrating extra support on peak days.
- Keep the historical no-queue result only as a benchmark, not as the main experimental model.
- Do not describe Carrier_A/B/C capacities as real observed company capacities; describe them as scenario carrier profiles.

- For final refinement, add weekday-specific JaamSim logic only if S3/S4 peak-day overtime scenarios must also be simulated; otherwise, clearly present S3/S4 as R-side operating-policy extensions and S0-S2 as the JaamSim-implemented capacity scenarios.

9.2 Conclusions

The JaamSim implementation for S0-S2 confirms the same directional result as the R-side analysis: higher carrier capacity reduces mean total system time. The JaamSim means are moderately higher than the R-side benchmark values, with validation errors of about 7.5-7.8%, which is reasonable given the explicit queue-resource implementation. S3 and S4 remain useful R-side policy extensions, but their final JaamSim validation would require adding weekday-specific peak-day capacity logic.